

Design, Simulation and Manufacturing of an Optimized Connecting Rod using Autodesk Fusion 360

Cad Cam Project Report

Submitted By

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Abstract

The connecting rod is one of the most critical load-transmitting components in an internal combustion engine, continuously subjected to complex fluctuating tensile, compressive, and bending forces during engine operation. Due to these severe cyclic loading conditions, the component requires an optimal combination of high strength, stiffness, fatigue resistance, dimensional stability, and lightweight characteristics. The primary objective of this project was to design, analyze, optimize, and virtually manufacture a structurally efficient connecting rod using Autodesk Fusion 360 by integrating CAD, CAE, CAM, and generative design methodologies within a unified engineering workflow. Initially, a fully parametric three-dimensional CAD model of the connecting rod was developed using Fusion 360. The geometry was designed considering critical engineering aspects such as load transfer efficiency, manufacturability, stress concentration reduction, geometric symmetry, machining feasibility, and weight optimization. Multiple parametric dimensions including shank thickness, web geometry, fillet radius, bore dimensions, and pocket depth were controlled through user-defined parameters to facilitate rapid design iterations and optimization studies. Finite Element Analysis (FEA) was subsequently performed under tensile and compressive loading conditions to evaluate the structural behavior of the connecting rod. Important performance indicators such as Von Mises stress, total deformation, safety factor, and buckling resistance were investigated systematically for all design iterations. Five different design configurations were analyzed and compared quantitatively to study the effect of material removal and geometric modification on structural performance. The optimized design achieved approximately 48% mass reduction compared to the initial baseline model while maintaining acceptable structural safety and stiffness requirements. Advanced generative design techniques available within Fusion 360 were further utilized to automatically generate AI-assisted lightweight geometries based on load paths, preserved regions, manufacturing constraints, and material distribution optimization. The generated design demonstrated efficient material utilization and improved load transfer characteristics while significantly reducing unnecessary mass. The optimized connecting rod geometry was then manufactured virtually using Fusion 360 Manufacturing Workspace. A complete 3-axis CNC machining strategy consisting of adaptive roughing, contour finishing, pocket machining, parallel finishing, drilling, and boring operations was developed.

1. Objectives

The major objectives of the project are listed below:

1. To design a fully parametric connecting rod model using Autodesk Fusion 360.
2. To develop a lightweight and structurally efficient connecting rod geometry.
3. To perform Finite Element Analysis under tensile and compressive loading conditions.
4. To evaluate Von Mises stress, deformation, safety factor, and buckling characteristics.
5. To perform multiple design iterations for optimization.
6. To compare the performance of all design iterations quantitatively.
7. To apply generative design techniques for advanced topology optimization.
8. To develop a complete 3-axis CNC machining strategy using Fusion 360 CAM workspace.
9. To generate gouge-free toolpaths for roughing, finishing, pocketing, and bore machining.
10. To simulate machining operations and generate CNC-compatible G-code.

2. Introduction

A connecting rod is a vital mechanical component used in internal combustion engines to connect the piston to the crankshaft. It converts the reciprocating motion of the piston into rotary motion at the crankshaft. During engine operation, the connecting rod is subjected to highly fluctuating tensile and compressive forces generated due to combustion pressure and inertia effects.

The design of connecting rods requires careful consideration because the component must possess:

- High strength
- Good fatigue resistance
- Low weight
- High stiffness
- Good manufacturability

Reducing the mass of a connecting rod improves engine efficiency, reduces inertia forces, decreases vibration, and enhances fuel economy. However, excessive weight reduction may lead to high stress concentration, deformation, and buckling failure.

Modern engineering industries therefore utilize CAD, Finite Element Analysis (FEA), topology optimization, and generative design techniques to develop highly optimized mechanical components.

In this project, Autodesk Fusion 360 was used as the integrated platform for:

- CAD modeling
- Parametric design
- FEA simulation
- Generative design
- CAM toolpath generation
- CNC manufacturing simulation

A fully parametric connecting rod model was first developed and then analyzed under tensile and compressive loading conditions. Multiple design iterations were generated systematically to improve the strength-to-weight ratio. Generative design techniques were also used to automatically optimize material distribution along load paths.

Finally, the optimized connecting rod was manufactured virtually using Fusion 360 CAM workspace through complete CNC machining simulation and G-code generation.

The project demonstrates the practical implementation of modern digital engineering workflows for mechanical product development.

3. Methodology

3.1. CAD Modeling

The CAD model of the connecting rod was developed using Autodesk Fusion 360 by adopting a fully parametric and feature-based modeling approach. The design methodology focused on achieving an optimal balance between structural strength, lightweight performance, manufacturability, and geometric symmetry. The connecting rod geometry primarily consisted of the big-end bore, small-end bore, shank region, side pockets, and smooth filleted transition zones. The big-end bore was designed to accommodate the crankshaft journal, while the small-end bore was designed for piston pin assembly. The shank region was developed using an I-beam cross-sectional configuration due to its superior stiffness-to-weight characteristics and efficient bending resistance. Special attention was given to the transition regions between the bores and the shank section because these areas experience high stress concentration during operation. Smooth fillet radii

were introduced to minimize abrupt geometric discontinuities and improve load transfer characteristics throughout the structure.

The connecting rod consists of:

- Big-end bore
- Small-end bore
- I-beam shank section
- Side relief pockets
- Filleted transition regions

The big-end bore diameter was maintained at 30 mm and the small-end bore diameter at 15 mm. Smooth fillet transitions were incorporated to reduce stress concentration.

The geometry was designed symmetrically to ensure balanced load transfer and simplified manufacturability.

3.2. Parametric Design Methodology

Key dimensions were controlled using user-defined parameters. This enabled rapid design modification during optimization studies.

The important parameters included:

- Big-end diameter
- Small-end diameter
- Center distance
- Shank thickness
- Web thickness
- Pocket depth
- Fillet radius

Mathematical relationships were also introduced between dimensions to maintain geometric consistency.

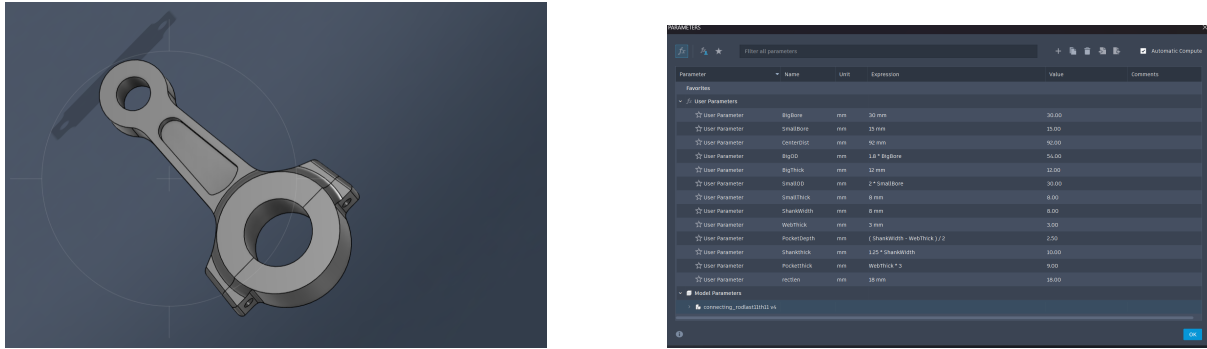


Figure 1: Geometric relationships and dimensional constraints used in the model

3.3. Finite Element Analysis

Finite Element Analysis was carried out using Autodesk Fusion 360 Simulation Workspace to evaluate the structural integrity and performance of the connecting rod under realistic loading conditions. The primary objective of the analysis was to study stress distribution, deformation characteristics, safety factor variation, and buckling resistance for different design iterations. Two major loading conditions were considered: 1. Tensile loading condition 2. Compressive loading condition For the tensile load case, the small-end bore was constrained while an 8 kN tensile load was applied at the big-end bore. This condition simulated the tensile forces generated during piston acceleration and high-speed engine operation. The analysis primarily focused on determining Von Mises stress distribution and total deformation behavior. For the compressive load case, a 12 kN compressive force was applied to evaluate buckling resistance and compressive structural stability. Since connecting rods are slender structural members, buckling behavior becomes highly critical during compressive loading conditions generated by combustion pressure

3.3.1 Tensile Load Case

- Small-end fixed
- 8 kN tensile load applied at big-end
- Objective: determine stress and deformation

3.3.2 Compressive Load Case

- Small-end fixed
- 12 kN compressive load applied
- Objective: evaluate buckling resistance

The following results were obtained:

- Von Mises stress

- Total deformation
- Safety factor
- Buckling load factor

3.4. Design Iteration Process

Five successive design iterations were developed.

Table 1: Design Iteration Strategy

Iteration	Objective
1	Baseline heavy design
2	Initial weight reduction
3	Aggressive lightweight design
4	Balanced manual optimization
5	Generative optimized design

The geometry was modified by changing:

- Shank thickness
- Web geometry
- Pocket depth
- Fillet transitions
- Load transfer regions

3.5. Generative Design

Fusion 360 Generative Design workspace was used to develop AI-assisted optimized geometry.

The software automatically redistributed material based on:

- Load paths
- Boundary conditions
- Preserved regions
- Manufacturing constraints

The generated design achieved significant weight reduction while maintaining acceptable structural behavior.

3.6. CAM Toolpath Generation

The optimized connecting rod was manufactured virtually using Fusion 360 Manufacturing workspace.

A complete 3-axis CNC machining strategy was developed involving:

- Adaptive roughing
- Pocket machining
- Contour finishing
- Parallel finishing
- Scallop finishing
- Drilling
- Boring cycles

Three separate machining setups were used to machine all surfaces efficiently.

3.7. Simulation and G-code Generation

Toolpath simulations were performed to verify:

- Collision-free machining
- Gouge-free machining
- Proper cutter engagement
- Smooth machining sequence

Finally, CNC-compatible G-code was generated successfully.

4. Results and Discussion

4.1. Comparative Iteration Results

Table 2: Comparison of All Design Iterations

Iteration	Mass (g)	Max Stress (MPa)	Safety Factor	Buckling Factor
1	791.132	40.7	6.24	97.7
2	515.291	74.66	3.40	19.4
3	442.053	114.78	2.21	8.48
4	523.463	55.50	4.58	20.02
5	411.162	105.84	2.40	1.77

4.2. Iteration 1 Analysis

The first iteration represented a conservative heavy design with very high stiffness and excellent buckling resistance. However, the mass was excessively high and material utilization was inefficient.

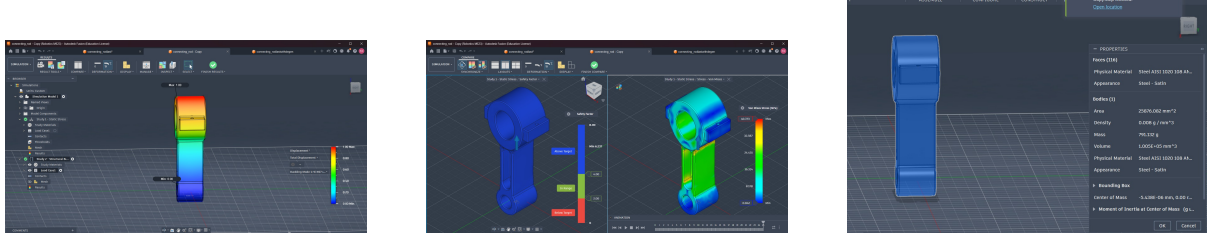


Figure 2: Iteration 1 CAD model, stress distribution, and deformation results

4.3. Iteration 2 Analysis

The second iteration focused on weight reduction by modifying the shank and web regions. Significant mass reduction was achieved while maintaining acceptable safety factor.

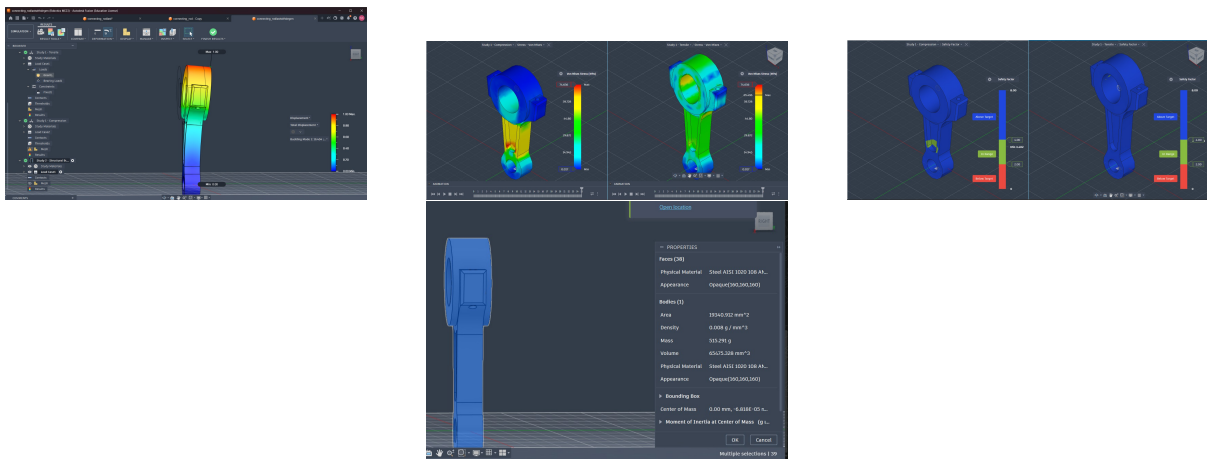


Figure 3: Iteration 2 CAD model, stress distribution, and deformation results

4.4. Iteration 3 Analysis

Aggressive material removal reduced mass further but resulted in high stress concentration and reduced buckling resistance

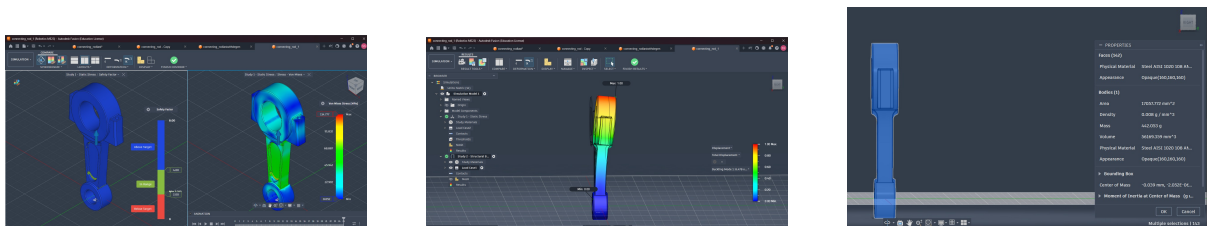


Figure 4: Iteration 3 CAD model, stress distribution, and deformation results

4.5. Iteration 4 Analysis

Iteration 4 represented the best manually optimized conventional design. It achieved balanced performance between:

- Structural safety
- Manufacturability
- Weight reduction
- Buckling resistance

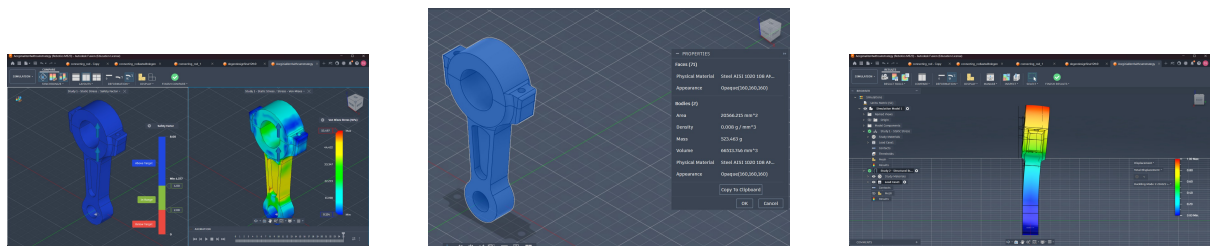


Figure 5: Iteration 4 CAD model, stress distribution, and deformation results

4.6. Iteration 5 Analysis

The fifth iteration used generative design optimization.

Advantages included:

- Lowest mass
- Efficient material distribution
- Improved load path geometry

However, slender geometry reduced buckling stability

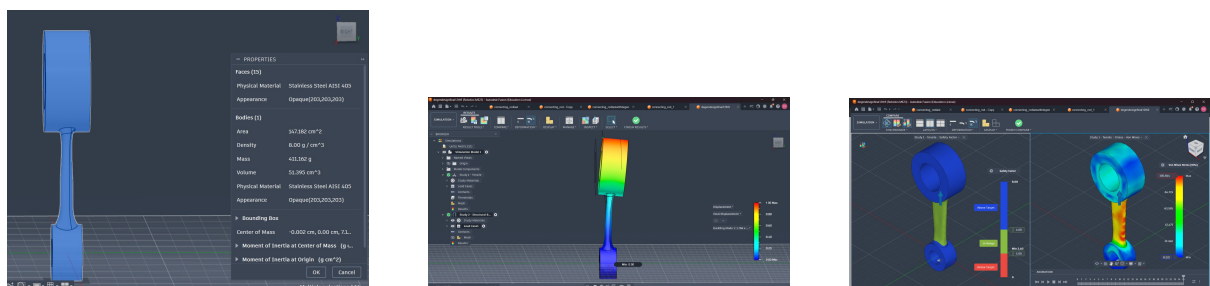


Figure 6: Iteration 5 CAD model, stress distribution, and deformation results

4.7. Mass Reduction Analysis

The total mass reduction achieved from Iteration 1 to Iteration 5 was approximately:

$$\text{Mass Reduction} \approx 48\%$$

This demonstrated the effectiveness of FEA-driven optimization and generative design methodologies.

4.8. Stress Distribution Analysis

Critical stress regions were consistently observed near:

- Big-end transitions
- Fillet regions
- Pocket termination zones
- Bore transitions

Smoother load transfer geometry significantly improved stress distribution in later iterations.

5. CAM Toolpath Generation and Manufacturing Simulation

The optimized connecting rod was manufactured virtually using Autodesk Fusion 360 Manufacturing Workspace by developing a complete 3-axis CNC machining strategy. The CAM process focused on achieving efficient material removal, smooth surface finish, accurate bore machining, and collision-free tool movement while minimizing machining time. A DMG MORI CMX 600 V vertical machining center was selected for the virtual manufacturing process due to its high precision, rigidity, and compatibility with multi-operation CNC machining. Three separate machining setups were developed to provide proper accessibility to all critical surfaces and ensure accurate machining of the front side, back side, and bore regions. The first setup primarily involved roughing and finishing operations on the front side of the component. Adaptive clearing strategies were used for high-efficiency roughing because adaptive toolpaths maintain constant cutter engagement and reduce sudden tool loading, thereby improving machining stability and tool life.

5.1. Machine Setup

Machine Used:

- DMG MORI CMX 600 V
- 3-axis CNC vertical machining center

Three machining setups were created:

Table 3: Machining Setup Strategy

Setup	Orientation	Purpose
1	Front Side	Roughing and finishing
2	Back Side	Pocket machining and finishing
3	Final Setup	Bore finishing and contouring

5.2. CAM Operations

The following operations were implemented:

- Face milling
- Adaptive clearing
- 2D pocket machining
- Contour finishing
- Parallel finishing
- Scallop finishing
- Drilling cycles
- Boring cycles

5.3. Machining Time Records

Table 4: Estimated Machining Times

Setup	Machining Time
Setup 1	17 min 23 sec
Setup 2	2 hr 03 min 54 sec
Setup 3	2 hr 14 min 55 sec

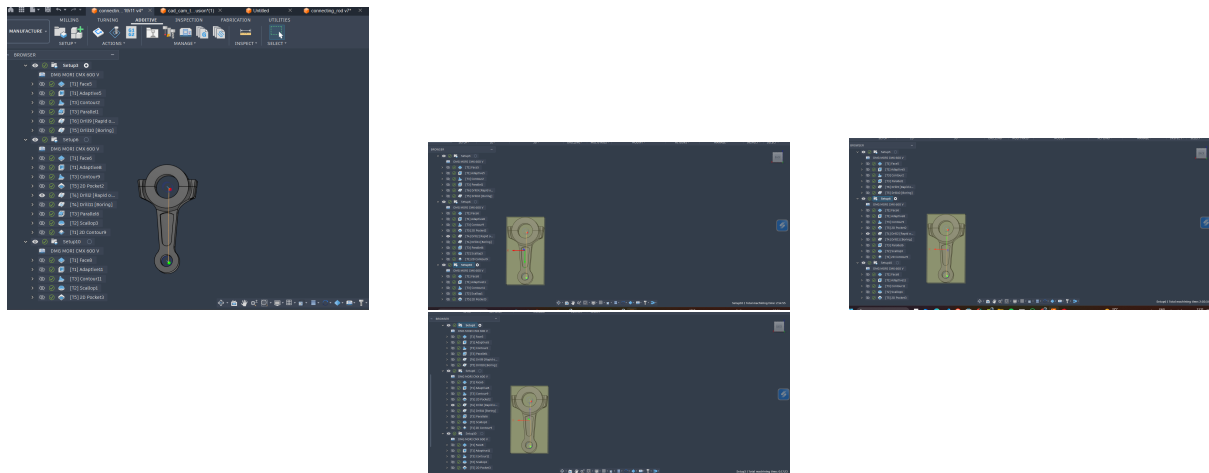


Figure 7: Cam tool path time for different setups

5.4. Toolpath Verification

Fusion 360 simulation confirmed:

- Collision-free machining
- Gouge-free toolpaths
- Proper cutter engagement
- Smooth machining sequence

No interference between tool, holder, fixture, or workpiece was observed.

5.5. Sample G-code Header

```
%
O1001
(Toolpath: Connecting Rod Machining)
(Machine: DMG MORI CMX 600 V)
G90 G94 G17
G21
G28 G91 Z0
T1 M06
S2500 M03
G54
G00 X0 Y0
G43 Z50 H01
M08
#The total code is in the file
```

6. Conclusion

The project successfully demonstrated the complete engineering workflow involved in the design, analysis, optimization, and virtual manufacturing of a connecting rod using Autodesk Fusion 360. The integration of CAD modeling, Finite Element Analysis, generative design, CAM toolpath generation, and CNC machining simulation enabled the development of a structurally efficient and lightweight mechanical component. A fully parametric connecting rod model was developed systematically and optimized through multiple design iterations. Structural simulations under tensile and compressive loading conditions helped identify critical stress concentration regions, deformation characteristics, and buckling behavior. The iterative optimization process enabled significant reduction in component mass while maintaining acceptable safety factor and structural reliability. The final optimized design achieved nearly 48% mass reduction from the baseline configuration, demonstrating the effectiveness of simulation-driven optimization and intelligent material distribution methodologies. Generative design further enhanced structural efficiency by automatically redistributing material along critical load paths.

References

- [1] Autodesk Fusion 360 Documentation, Autodesk Inc., 2025.
- [2] R. G. Budynas and J. K. Nisbett, *Shigley's Mechanical Engineering Design*, 11th Edition, McGraw Hill, 2020.
- [3] S. S. Rao, *The Finite Element Method in Engineering*, 6th Edition, Butterworth-Heinemann, 2017.
- [4] M. P. Groover, *Fundamentals of Modern Manufacturing*, Wiley Publications, 2021.
- [5] J. T. Black and R. A. Kohser, *DeGarmo's Materials and Processes in Manufacturing*, Wiley, 2020.
- [6] Fusion 360 CAM User Guide, Autodesk Manufacturing Solutions, 2025.
- [7] A. Esawi and M. Ashby, "Computer-based selection of materials and processes," *Materials and Design*, vol. 24, no. 6, pp. 479–484, 2003.